# General Chemistry Foundations

Ingyu Bahng

July 2026

## Contents

# 1 Acids and Bases

## 1.1 Acid-Base Basics

**Core Concept 1.1: Acids**

**Acids** are substances that increase the $[H^+]$ of a solution.

(shorthand) $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

(more accurately) $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

**Core Concept 1.2: Bases**

**Bases** are substances that decrease the $[H^+]$ of a solution (usually through the addition of $OH^-$, however there are exceptions).

(shorthand) $BOH(aq) + H^+(aq) \rightleftharpoons B^+(aq) + H_2O(aq)$

(more accurately) $BOH(aq) + H_3O^+(aq) \rightleftharpoons B^+(aq) + 2\,H_2O(l)$

**Core Concept 1.3: Brønsted-Lowry Theory**

**Brønsted-Lowry Theory** defines acids as $H^+$ donors and bases as $H^+$ acceptors (not just $OH^-$ donors).

As covered in Section 2, acid-base reactions also have equilibriums (*denoted as* $K_{a/b}$) specific to each acid/base.

**Core Concept 1.4: Strong/Weak Acids and Bases**

Acid-base reactions with high $K_{a/b}$ are categorized as **strong acids/bases** ($K_{a/b} >> 1$) and those with low $K_{a/b}$ as **weak acids/bases** ($K_{a/b} << 1$). Strong acids and bases also tend to conduct electricity better because of more dissociated ions.

**Core Concept 1.5: Indicators**

**Indicators** (In) are weak acids or bases that change color at some $[H^+]$ in a solution. The color change occurs because the corresponding acid (HIn) and base ($In^-$) forms of the indicators possess different colors.

$HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq)$

thus $K_{In}$ is defined as

$$K_{In} = \frac{[H^+][In^-]}{[HIn]}$$

Here are a two examples of indicators that are commonly used in the lab setting.

Anthocyanin: $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

Anthoxanthin: $HB(aq) \rightleftharpoons H^+(aq) + B^-(aq)$

Anthocyanin shows either a red or blue hue based on the comparative level of its acid form to its corresponding base. Anthoxanthin, on the other hand, is either clear or yellow. Through Le Chatelier's principle, we can manipulate the color of the indicator solution by adding a strong acid or base to raise or lower $[H^+]$, respectively.

## 1.2 Acid-Base Stoichiometry

**Core Concept 1.6: Amphiproticity**

Water is **amphiprotic**, meaning that it can both accept or donate an $H^+$.

($H^+$ Acceptor) $HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$

($H^+$ Donor) $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

Water also maintains equilibrium with the autoionized form of itself. Heat is a reactant in this autoionization reaction, thus an increase or decrease in temperature will shift the reaction right and left, respectively.

$H_2O(l) + H_2O(l) + \text{heat} \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

Therefore the equlibrium of $H_2O(l)$ is represented as

$$K_W = [H3O^+][OH^-] = [H^+][OH^-]$$

**Core Concept 1.7: pH**

**pH** is a quantitative measure of how acidic (*low pH*) or basic (*high pH*) a solution is. Mathematically, pH is defined as

$$pH = -\log_{10}[H^+]$$

| Water temperature (°C) | $K_w$ | $[H^+]$ | pH |
|---|---|---|---|
| 0 | $10^{-14.94}$ | $10^{-7.47}$ | 7.47 |
| 5 | $10^{-14.73}$ | $10^{-7.36}$ | 7.36 |
| 10 | $10^{-14.53}$ | $10^{-7.26}$ | 7.26 |
| 15 | $10^{-14.35}$ | $10^{-7.18}$ | 7.18 |
| 20 | $10^{-14.17}$ | $10^{-7.08}$ | 7.08 |
| 25 | $10^{-14.00}$ | $10^{-7.00}$ | 7.00 |
| 30 | $10^{-13.83}$ | $10^{-6.92}$ | 6.92 |
| 35 | $10^{-13.68}$ | $10^{-6.84}$ | 6.84 |
| 40 | $10^{-13.53}$ | $10^{-6.76}$ | 6.76 |

Table 1: Water pH across different temperatures

A common misconception people often assume is that the pH of 7 corresponds to a neutral solution of water ( $[H^+] = [OH^-]$ ). From Table 1 however, we see that **the pH of 7 being neutral only applies to a temperature of 25°C**. At higher temperatures, the equilibrium will shift right to produce more $H_3O^+$, effectively making a lower pH value as the new neutral base point.

From a molecular perspective, higher temperatures increases autoionization rates because the higher kinetic energy of the water molecules makes it statistically more likely to collide/react with one another in order to form $H_3O^+$ and $OH^-$, thus decreasing the pH of the solution.

Let's take a moment to look at stoichiometric calculations for acids and bases.

**Core Concept 1.8: Strong Acid/Base Stoichiometry**

For **strong acids and bases**, we rely on simple stoichiometry to calculate $[H^+]$. We'll use HCl as an example.

$HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$

The equilibrium for HCl is represented as

$$K_a = \frac{[H_3O^+][Cl^-]}{[HCl]} >> 1$$

The $K_{a/b}$ for strong acids and bases are generally very large values. The approximate $K_a$ for HCl is $10^7$. Thus when dealing with strong acids and bases, we can assume the initial [HCl] to be equal to the final $[H3O^+]$ with a negligible margin of error.

$$[HCl] = [H_3O^+] = [H^+]$$

Subsequently, you can calculate pH using the formala $pH = -\log_{10}[H^+]$

**Core Concept 1.9: Weak Acid/Base Stoichiometry**

For **weak acids and bases**, we rely on more complex stoichiometry to calculate $[H^+]$. We'll use acetic acid ($CH_3COOH$) as an example.

$CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$

The equilibrium for acetic acid is represented as

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} << 1$$

The $K_{a/b}$ for weak acids and bases are generally very small values. The approximate $K_a$ for acetic acids is $1.8 \times 10^{-5}$. Thus when representing the $K_a$ of acetic acid, we can derive the equation

$$K_a = 1.8 \times 10^{-5} = \frac{[H_3O^+]_{final}[CH_3COO^-]_{final}}{[CH_3COOH]_{initial} - [H_3O^+]_{final}} \approx \frac{[H_3O^+]_{final}[CH_3COO^-]_{final}}{[CH_3COOH]_{initial}}$$

Here, since a mole of $CH_3COOH$ is consumed every mole of $H_3O^+$ or $CH_3COO^-$ is formed, $[CH_3COOH]_{final}$ must be equal to $[CH_3COOH]_{initial} - [H_3O^+]_{final}$. However, since $K_a$ is extremely small, we approximate $[CH_3COOH]_{final}$ as just $[CH_3COOH]_{initial}$ because $[H_3O^+]_{final}$ value is negligible in comparison to $[CH_3COOH]_{initial}$.

The smaller the $K_{a/b}$ of a weak acid or base is, the more accurate the stoichiometric calculation will be.

**Core Concept 1.10: Conjugate Acid-Base Pairs**

The corresponding base that is formed when an acid donates a $H^+$ is called the **conjugate base**. Likewise, the corresponding acid that forms when a base accepts a $H^+$ is called the **conjugate acid**.

For example, the conjugate acid for $NH_3$ is $NH_4^+$, and the conjugate base for $HNO_3$ is $NO_3^-$. These are represented as

$NH_3(aq) + H^+(aq) \rightleftharpoons NH_4^+(aq)$

$HNO_3(aq) \rightleftharpoons H^+(aq) + NO_3^-(aq)$

Because the chemical reaction for an acid is the same as that of the conjugate base (*just reversed*), the $K_a$ for an acid is the inverse of the $K_b$ for its conjugate base. As such, the stronger the acid/base, the weaker the corresponding conjugate base/acid will be.

**Core Concept 1.11: Conjugate Acid-Base Pair Equilibriums and $K_w$**

The product of the equilibrium constants of an acid $K_a$ and its conjugate base $K_b$ (*or base and its conjugate acid*) is equal to the equilibrium constant of the autoionization of water $K_w$. We can derive this relationship with the following algebraic manipulations using the formulas for the shell reactions below.

(Acid) $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

(Conjugate Base) $A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq)$

Thus the equilibrium constants are $K_a = \frac{[H_3O^+][A^-]}{[HA]}$, $K_b = \frac{[HA][OH^-]}{[A^-]}$. When we rewrite $K_b$ as

$$K_b = \frac{[HA][OH^-]}{[A^-]} \times \frac{[H_3O^+]}{[H_3O^+]} = K_w/K_a \Rightarrow K_w = K_aK_b$$

This relationship is true for all conjugate acid-base pairs.

## 1.3 Titration and Buffers

**Core Concept 1.12: Titration**

**Titration** is the process of adding an acid or base to bring it about to a level where occurs a change at a previously known pH (via indicators). This laboratory technique **allows chemists to determine the pH of a solution** by observing the amount of base or acid it takes to reach the target pH level.

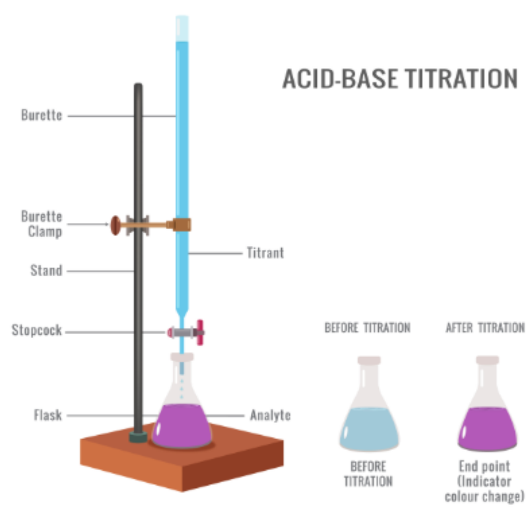


Figure 1: Basic titration set up

We call the solution that is being titrated the **analyte** and the solution that is doing the titrating the **titrant**.

**Core Concept 1.13: Strong Acid and Strong Base Titration**

An example of a **strong acid** (HCl) being titrated with a **strong base** (NaOH) is represented by the chemical equation $HCl(aq) + NaOH(aq) \rightleftharpoons NaCl(aq) + H_2O(l)$. Because of their high $K_{a/b}$ values, we can simplify this equation to the following.

$$H^+(aq) + OH^-(aq) \rightleftharpoons H_2O(l)$$

To neutralize a solution of an unknown [HCl], we would need to add equal amounts of NaOH. Thus, based on the amount of NaOH we add to the solution in order to reach this neutral point, we can calculate the original [HCl].

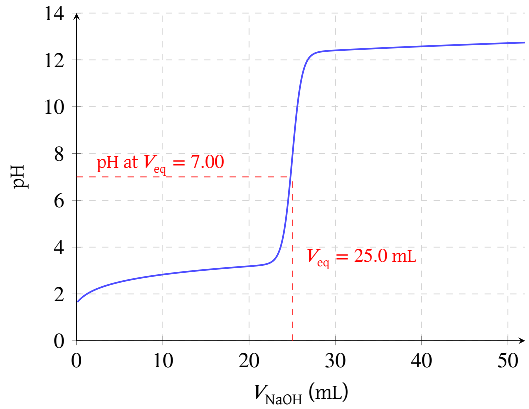


Figure 2: Strong acid-base titration curve

Figure 2 is a graphical representation of the analyte's pH as more titrant is added. The point at which the moles of titrant added is stoichiometrically equal to the amount of the analyte originally present is called the **equivalence point**. For the case of a strong acid with a strong base, this point is at a neutral pH of 7 at 25°C.

**Core Concept 1.14: Weak Acid/Base and Strong Base/Acid Titration**

The titration curve of a **weak acid** ($CH_3COOH$) being titrated by a **strong base** (NaOH) is shown below.

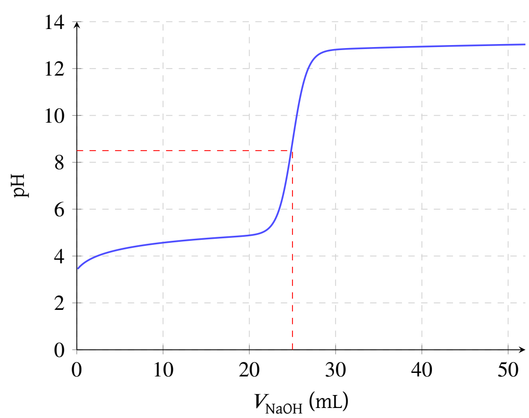


Figure 3: Weak acid and strong base titration curve

The equivalence point for this case **lies above the neutral pH**, indicating a basic state at equal parts weak acid and strong base. This is because once all $H^+$ is neutralized by the presence of $OH^-$ from NaOH, **what is left is the strong conjugate base molecule** $CH_3COO^-$. In the presence of this strong conjugate base, some water molecules will react with the $CH_3COO^-$ to form $CH_3COOH$ and $OH^-$, thus resulting in a higher pH value at equivalence. When confused, it's helpful to (1) assume the reaction will go to completion and then (2) look at what is left and ask how it affects the pH.

**Core Concept 1.15: Buffers**

**Buffers** are solutions containing both a weak acid and its corresponding conjugate base. They are used to **resist** large changes of pH upon addition of strong acids or bases.

Based on the chemical equation of a weak acid ($HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$), we can manipulate the $K_a$ equation to give us the final following formula.

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = [H_3O^+]\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

$$-\log K_a = -\log[H_3O^+] + \log\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

$$pH = pK_a + \log\frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

The final equation is a simple way to determine the pH of a buffer solution under a given concentration of the weak acid and its conjugate base. *Notice how the pH of a solution will be equivalent to the $pK_a$ when there are equal parts conjugate acid and base*

The making a buffer solution can be done in one of three ways.

(a) Equal parts of the weak acid and its conjugate base

(b) Weak acid with half the amount of a strong base

(c) Conjugate base with half the amount of a strong acid

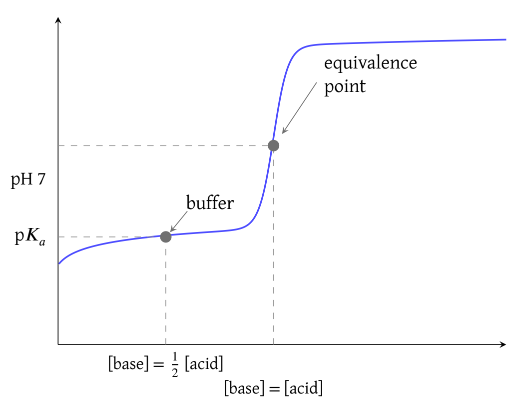


Figure 4: Buffer solution titration curve

Based on Figure 4 the $pK_a$ is also half way between the equivalence point and y-axis of the titration curve, the point at which [conjugate acid] = [conjugate base]. Any subsequent addition of a strong acid or base will shift the pH value, but the pH will be resistant to change within a given range until larger amounts of a strong acid or base is added.

Using this technique, chemists can customize the pH they want a solution to linger around based on the $K_a$ value of a weak acid. Conversely, a buffering solution can also be held at a basic pH level by adding equal parts of weak base with its corresponding strong conjugate acid.

# 2 Equilibrium

## 2.1 Reaction Rate & Equilibrium

**Core Concept 2.1: Reaction Rates**

Atoms, ions, and molecules react with one another via collisions. The more frequent and energetic the collisions, the faster the reaction rate.

Take the formation of compound C as an example: $A + B \longrightarrow C$. The rate at which C forms depends on factors such as [A], [B], temperature, the presence of a catalyst, etc. and can be expressed mathematically as:

$$\text{Rate} = k[A]^m[B]^n$$

Here, k is the **rate constant**, which is specific to the given reaction and its conditions. $m$ and $n$ are the **partial reaction orders**, which represents a reactant's influence over the reaction rate. Figure 5 is a hypothetical plot of [A], [B], and [C] over time.

In single-step reactions, partial reaction orders are equal to the stoichiometric coefficients of in the balanced chemical equation. However, in multi-step reactions, partial reaction orders may not directly correspond to the stoichiometric coefficients and must be determined experimentally.

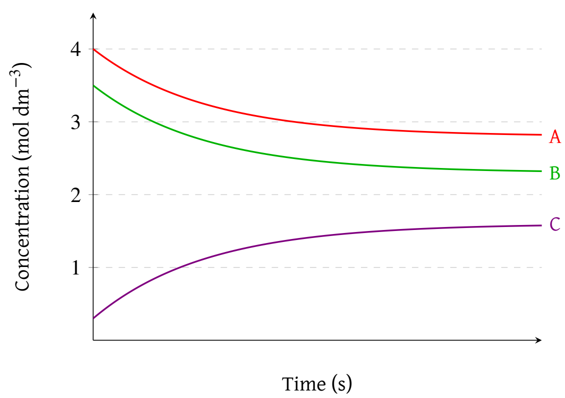


Figure 5: Equilibrium of $A + B \longrightarrow C$

Consider the decomposition of $H_2O_2$ ($H_2O_2 \longrightarrow H_2O + O_2$) as an example. When you graph the concentration of $H_2O_2$ over time, you get an exponential decay curve, where the demposition rate of $H_2O_2$ decreases as $[H_2O_2]$ decreases.

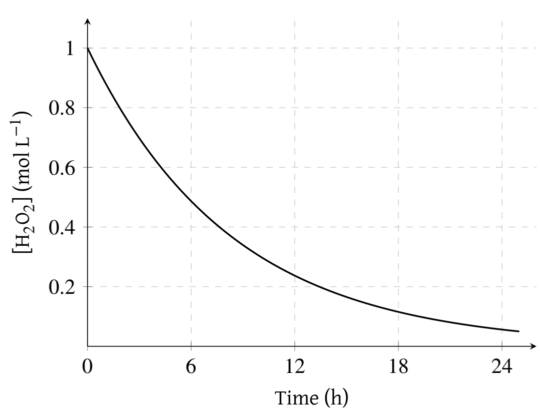


Figure 6: Decomposition of $H_2O_2$ over time

**Core Concept 2.2: Equilibrium**

Reactions do not always proceed in only one direction. Many reactions are **reversible**, meaning that the products can react to reform the reactants. When the rates of the forward and reverse reactions are equal, the system is said to be at **equilibrium**. Thus, for the general reversible shell reaction $AB \rightleftharpoons A + B$, the Eq reaction state can be expressed as:

$$k_{forward}[AB] = k_{backward}[A][B]$$

And the equilibrium constant $K_c$ is defined as:

$$K_c = \frac{k_f}{k_b} = \frac{[A][B]}{[AB]}$$

**Core Concept 2.3: Analyzing $K_c$**

$K_c$ provides insight into the position of equilibrium. A large $K_c$ (>> 1) indicates product favorability, while a small $K_c$ (<< 1) indicates reactant favorability.

Some examples of reactions that possess a **large** $K_c$ (reaction to completion) include:

$Si + O_2 \longrightarrow SiO_2 \quad K_c \approx 10^{150}$

$2\,H_2 + O_2 \longrightarrow H_2O \quad K_c \approx 10^{81}$

Some examples of reactions that possess a **moderate** $K_c$ (half-reaction) include:

$CO + \text{Hemoglobin} \longrightarrow CO \cdot \text{Hemoglobin} \quad K_c \approx 300$

$O_2 + \text{Hemoglobin} \longrightarrow O_2 \cdot \text{Hemoglobin} \quad K_c \approx 4$

$N_2O_4 \longrightarrow 2\,NO_2 \quad K_c \approx 0.2$

An example of a reaction that possess a **neglible** $K_c$ (no reaction) include:

$N_2 + O_2 \longrightarrow 2\,NO \quad K_c \approx 10^{-31}$

The decomposition of Bromothymol Blue (*denoted as A*) exhibits a nice visual representation of reaction equilibria.

$HA \rightleftharpoons H^+ + A-$

If we start with a pure HA solution, the solution will gradually change color from yellow to green to blue as the forward reaction proceeds and $[A^-]$ rises. This particular compound is also known as an indicator, a concept covered in Core Concept 1.5.

**Core Concept 2.4: Le Chatelier's Principle**

**Le Chatelier's Principle** states that a system in equilibrium subjected to a stress will react in a direction that tends to reduce that stress and return back to Eq.

In more exact terms, stress will shift the **reaction quotient (Q)**—which represents the relative amounts of reactant and products at a given point in time before it has nessecarily reached equilibrium ($Q \neq K$)—and the system will react in a manner that shifts Q back to K.

$$Q = \frac{[A]_{current}[B]_{current}}{[AB]_{current}}$$

There are three major categories of stress in the context of Le Chatelier's Principle.

**Core Concept 2.5: Le Chatelier's Principle (*Concentration Change*)**

Shifting the point concentrations of any chemical participant will shift Q.

Let's reexamine the Bromothymol Blue reaction ($HA \rightleftharpoons H^+ + A^-$). If we add more $H^+$ to an initially green solution ( $[HA] = [A^-]$ ), the system will increase the backwards reaction rate to consume excess $H^+$, resulting in a more yellow solution. Conversely, removing $H^+$ (*by adding $OH^-$*) will increase the forward reaction rate to produce more $H^+$, resulting in a more blue solution.

Consider another example ($PCl_5 \longrightarrow PCl_3 + Cl_2$) based on the visual below. The plot shows the abrupt addition of $PCl_5$ to a system originally at equilibrium. Consequently, $Q < K$ and the system reacts by increasing the forwards reaction rate to re-establish equilibrium, resulting in a subsequent increase in $[PCl_3]$ and $[Cl_2]$.

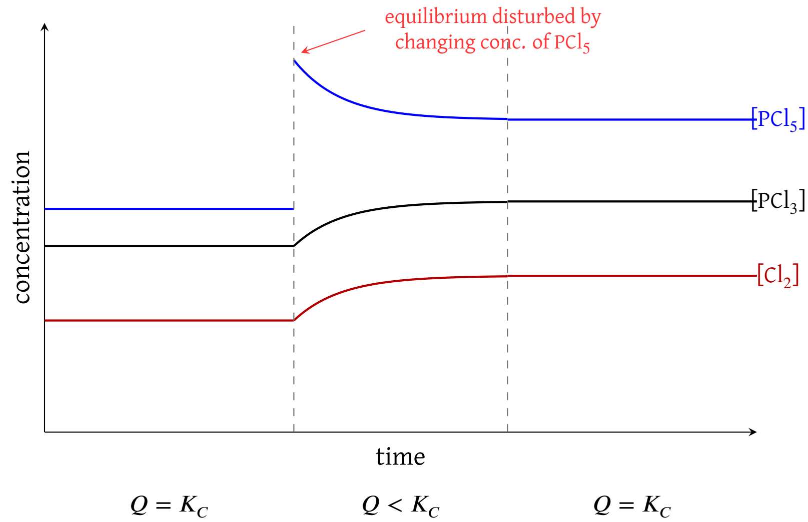


Figure 7: Disturbance of equilibrium by changing concentration of $PCl_5$

**Core Concept 2.6: Le Chatelier's Principle (*Pressure & Volume Change*)**

While concentration is the primary focus for solutions, pressure and volume are the key variables for gaseous systems. Increasing the pressure forces the equilibrium toward the side with fewer gas molecules to relieve the stress. Conversely, decreasing pressure allows the reaction to shift toward the side with more gas molecules.

The equilibrium constant in terms of partial pressures for gaseous reactions is written as $K_p$.

Consider the following reaction:

$$2\,NO_2(g) \rightleftharpoons N_2O_4(g) \qquad K_p = \frac{P_{N2O2}}{(P_{NO2})^2} \approx 7 \text{ at } 25°C$$

Increasing the volume ($V$) causes a drop in total pressure ($P$). To counteract this, the equilibrium shifts to the left, favoring the side with more moles of gas. Mathematically, this occurs because the partial pressure of $NO_2$ is squared in the denominator of the $Q_p$ expression. A decrease in pressure impacts the squared term more significantly than the linear numerator, making $Q < K$ and forcing a shift toward the reactants.

**Core Concept 2.7: Le Chatelier's Principle (*Temperature Change*)**

A temperature shift is analogous to an increase or decrease of heat within a system, and the Q shift direction depends on whether heat is a net product (**exothermic**) or a reactant (**endothermic**) of the chemical reaction at hand. Reconsider the following gaseous reaction in the context of heat.

$2\,NO_2(g) \rightleftharpoons N_2O_4(g) + \text{heat}$

Much like the addition of $N_2O_4$, increasing the temperature drives the equilibrium to the left to consume the excess energy. Conversely, cooling the system shifts the reaction to the right as it attempts to regenerate the lost heat.

Chemical equilibrium in the context of temperature is eleaborated more upon in Subsection 2.3.

## 2.2 Types of Equilibria

Equilibria can also be categorized into types.

**Core Concept 2.8: Homogeneous Equilibria**

In **homogeneous equilibria**, all reactants and products are in the same phase (solid, liquid, or gas), such as the following examples.

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons FeSCN^{2+}(aq)$ Solution

$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$ Gas

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$ Acid-Base

In **heterogeneous equilibria**, the reactants and products are in different phases and can be further subcategorized into three domains.

**Core Concept 2.9: Heterogeneous Equilibria (*Solubility of Solids*)**

The solubility of solids equilibrium refers to the dissociation and crystallization rate of solutes. Consider the dissolution of glucose. Solutes that do not dissociate are called **molecular or nonelectrolyte solutes**.

$C_6H_{12}O_6(s) \rightleftharpoons C_6H_{12}O_6(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[C_6H_{12}O_6(aq)]$

The dissociation rate only depends on the surface area (SA) of the solid glucose, while the crystallization rate depends on both the SA and $[C_6H_{12}O_6]$. At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [C_6H_{12}O_6(aq)]$$

For glucose, the $K_{sp}$ value is 5.1M at 25°C, meaning that the maximum $[C_6H_{12}O_6]$ in water is 5.1M. Beyond this concentration, any excess glucose will remain undissolved.

Now consider a different compound like NaCl. Solutes that dissociate are called **ionic or eletrolyte solutes**.

$NaCl(s) \rightleftharpoons Na^+(aq) + Cl^-(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[Na^+(aq)][Cl^-(aq)]$

At equilibrium:

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [Na^+(aq)][Cl^-(aq)]$$

At 25°C, the $K_{sp} \approx 37.3$, making the maximum [NaCl] in water approximately $6.11M$.

**Core Concept 2.10: Heterogeneous Equilibria (*Solubility of Gases*) & Henry's Law**

Gases can disolve into liquids just as how solids can. Consider the $K_{sp}$ of $CO_2$ in water.

$CO_2(g) \rightleftharpoons CO_2(aq)$

When we reach equilibrium, we have the following:

$$K_H = \frac{[CO_2(aq)]}{P_{CO2}} \rightarrow [CO_2(g)] = K_H P_{CO2}$$

The equilibrium formula above is known as **Henry's Law**, and the constant $K_H$ is called the **Henry's Constant** ($M/atm$), a quantitative measure of the proportion between a gas's solubility in a liquid and its partial pressure. The higher the $K_H$ value, the easier it is for a gas to dissolve in a liquid.

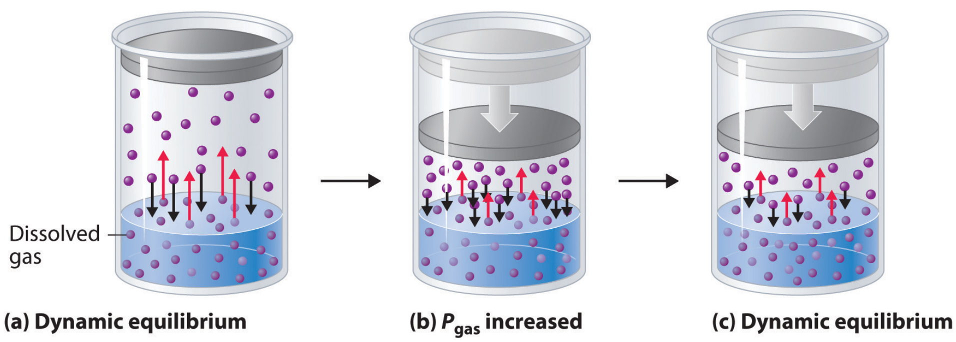


Figure 8: Gas solubility dynamic equilibrium

**Core Concept 2.11: Heterogeneous Equilibria (*Phase Change*)**

Phase changes also experience equilibriums. Consider the vaporization of water in a sealed flask.

$H_2O(l) \rightleftharpoons H_2O(g)$

Evaporation Rate = $k_f(SA)$

Condensation Rate = $k_c(SA)\, P_{H_2O}$

Just as the solubility equilibria for nonelectrolyte solutes is only dependent on the solvent concentration, the phase change equilibria for a gas to liquid transition is only dependent on $P_{H_2O}$.

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = P_{H_2O}$$

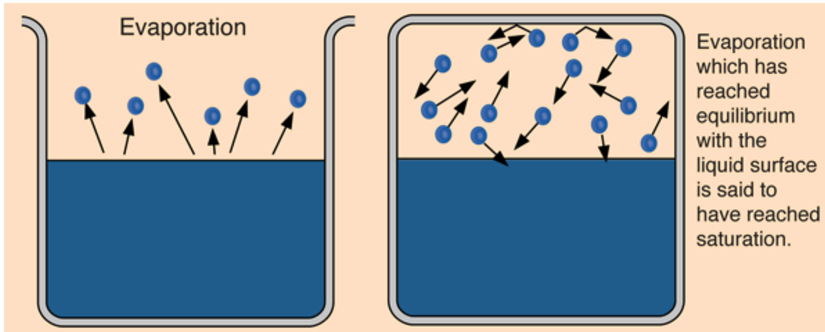


Figure 9: Phase change equilibrium of liquid-gas systems

*Phase change equilibria become irrelevant for solid-liquid transitions because both pure solids and pure liquids have fixed, constant concentrations that are excluded from the equilibrium expression, leaving no variable quantities and rendering the equilibrium constant meaningless.*

## 2.3 Shifting Equilibria

In **phase change equilibria**, the melting and boiling points of $H_2O$ at 0°C and 100°C respectively are defined at a standard pressure of 1 atm; under varying pressure conditions, these transition temperatures become dynamic.

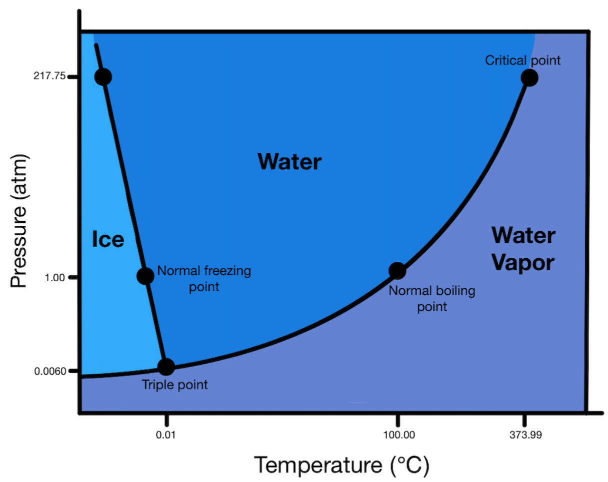


Figure 10: Phase diagram of water

Temperature shifts have a more nuanced effect in the context of equilibrium compared to the simple Le Chatelier's principle we have seen in Core Concept 2.7. Namely, the $K_{sp}$ values of various solutes change across temperature ranges.

On the molecular scale, $K_{sp}$ values of solid solutes increase with higher temperatures because the increased molecular motion of the solute and solvent molecules raises the probability of the solute entering the aqueous state.

In the context of gas solubility within a closed system, the Henry's Law constant $K_H$ decreases with increasing temperature. Since gaseous molecules possess greater kinetic energy than their aqueous counterparts, the dissolution of a gas is an exothermic process; as temperature rises, the equilibrium shifts to favour the gaseous state, increasing partial pressure $P$ at the expense of aqueous concentration $C$, thereby reducing $K_H = \frac{C}{P}$.

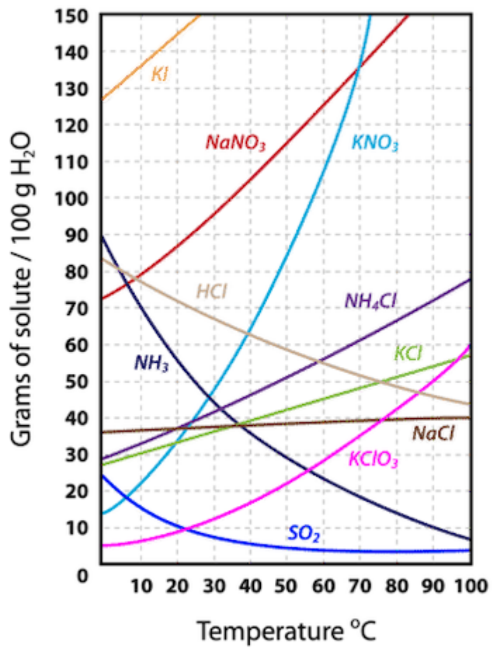


Figure 11: Solubility curve of various compounds

# 3 Gases

## 3.1 Kinetic Molecular Theory

**Core Concept 3.1: Kinetic Molecular Theory**

The **Kinetic Molecular Theory (KMT)** describes the behavior of gases based on the following postulates:

(a) Gas particles are in **constant, random motion** and travel in **straight lines** until they collide with another particle or the walls of the container.

(b) The particles themselves are so small compared to the space between them that their **individual volume is considered negligible** relative to the total volume of the gas.

(c) Particles **do not attract or repel one another**; there are no intermolecular forces between gas particles.

(d) Collisions between particles, and between particles and the walls of the container, are **perfectly elastic**, meaning no kinetic energy is lost during a collision.

(e) The **average kinetic energy** of the particles is **directly proportional to the absolute (Kelvin) temperature** of the gas, and all gases at the same temperature have the same average kinetic energy.

Its primary purpose is to provide a **theoretical model** that explains the macroscopic properties of matter (things we can measure, like pressure, volume, and temperature) based on the microscopic motion of individual atoms and molecules.

**Core Concept 3.2: Maxwell-Boltzmann Distribution**

The **Maxwell-Boltzmann distribution** describes the distribution of molecular speeds in a gas sample at a given temperature. As temperature increases, distribution broadens, indicating a wider range of speeds.

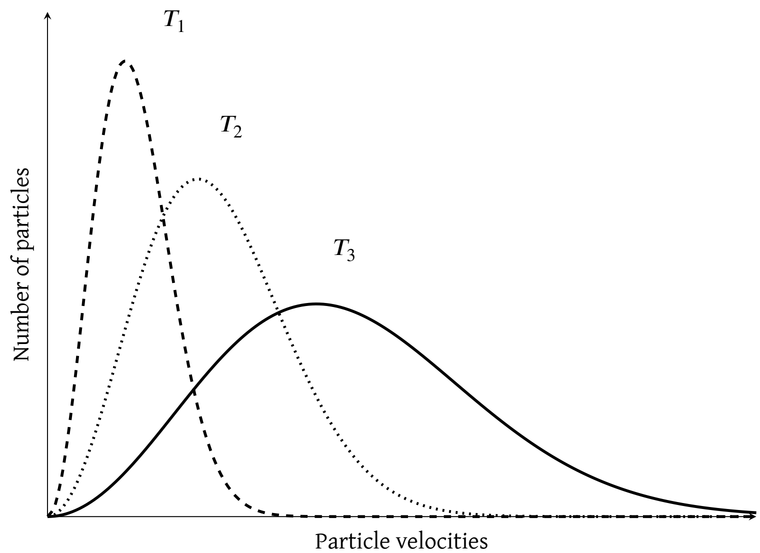


Figure 12: Maxwell-Boltzmann distribution (*particle velocities vs number of particles*)

Pressure (P) is the force per unit area (F/A) exerted by gas molecules colliding with the container walls, and depends on both the frequency and energy of those collisions. Consequently, an increase in T will either raise P through more frequent collisions, or increase V to maintain the same collision frequency—reflecting the **inverse relationship** between P and V. The proportional relationship between P and T is known as **Gay-Lussac's Law**, while the inverse relationship between V and P is known as **Boyle's Law**.

At the same temperature, molecules of different masses share the same average kinetic energy—meaning lighter molecules move faster than heavier ones. Consequently, the volume of an ideal gas is directly proportional to *n* regardless of the identity of the gas, which is known as **Avogadro's Law**.

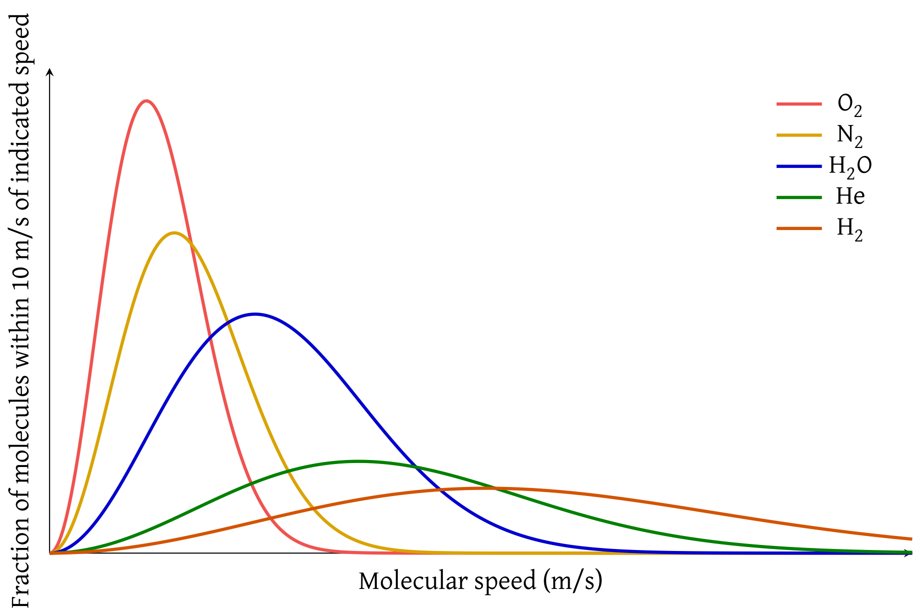


Figure 13: Molecular speeds of various molecules at the same temperature

**Core Concept 3.3: Ideal Gas Law & STP**

The relationships derived from the Maxwell-Boltzmann distribution—Gay-Lussac's Law ($P \propto T$), Boyle's Law ($P \propto \frac{1}{V}$), and Avogadro's Law ($V \propto n$)—can be unified into a single expression known as the **Ideal Gas Law**:

$$PV = nRT$$

where P is pressure, V is volume, n is the number of moles of gas, T is temperature in Kelvin, and R is the ideal gas constant, determined experimentally to be $0.0821\ \mathrm{L \cdot atm\ (mol \cdot K)^{-1}}$.

A useful reference point for ideal gas calculations is **Standard Temperature and Pressure (STP)**, defined as $T = 273\ \mathrm{K}$, $P = 1$ atm, under which one mole of an ideal gas occupies exactly 22.4 L.

## 3.2 Real Gases

**Core Concept 3.4: Real Gas Conditions**

No gas is truly ideal, and the Ideal Gas Law only approximates real gas behavior under certain conditions (e.g., low $P$ and high $T$). The two primary sources of deviation are:

(a) **Gas molecules have finite volume**—at high P, the volume occupied by the molecules themselves becomes significant relative to the total container volume, violating the ideal assumption of point-like particles.

(b) **Intermolecular forces (IMFs) exist between gas molecules**—at low T, the kinetic energy of molecules decreases, making attractive IMFs increasingly significant. When the average KE falls below the threshold needed to overcome these forces, molecules become captured and held together, transitioning first from gas to liquid (**condensation**) and eventually from liquid to solid as KE decreases further.

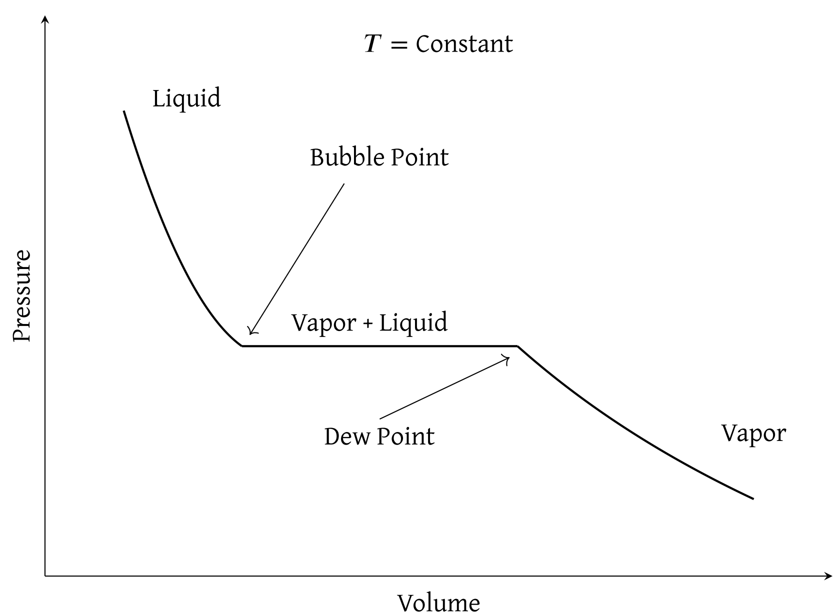


Figure 14: PV Isotherm Diagram

**Core Concept 3.5: Van der Waals Equation**

To account for the deviations of real gases, the **Van der Waals equation** modifies the Ideal Gas Law as follows:

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

The two correction terms each address one source of non-ideal behavior:

(a) $\frac{an^2}{V^2}$ corrects for **IMFs between gas molecules**. Molecules near the container wall are pulled inward by neighboring molecules, reducing the experimentally measured P below its true value. This term is therefore added to the measured P to recover the true pressure.

(b) $nb$ corrects for **the finite volume of gas molecules**, subtracting the volume physically occupied by the molecules themselves from the total container volume—an assumption the Ideal Gas Law neglects entirely.

The constants $a$ and $b$ are unique to each gas and must be determined experimentally.